Supplementary Information (Online Resource 6)

Article title: How Teams Decide on Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Worked illustration of oversight tradeoffs using MCDA

This supplement presents a retrospective illustration of Multi-Criteria Decision Analysis (MCDA) for comparing alternative human-oversight configurations. It demonstrates the calculation procedure and the influence of weighting assumptions; it does not validate RE4HOAI, establish the effectiveness of MCDA, or demonstrate improved safety in practice. Dubljević et al. (2021) developed and applied a Multi-Attribute Impact Assessment (MAIA) questionnaire to systematically capture expert opinions on 13 potential harms and 8 potential benefits arising from different Automatic Vehicle (AV) deployment scenarios. This questionnaire served as the foundation for a formal MCDA process, enabling a structured comparison of competing system requirements such as safety, accountability, efficiency, and public trust. Their findings demonstrated that regulated AV deployment models that incorporate human oversight (e.g., fleet ownership) were ethically and socially preferable to fully autonomous or unregulated alternatives.

MCDA can structure comparisons among alternatives by making criteria, scores, and weighting assumptions explicit. In RE4HOAI, it is proposed as one possible support for stakeholder deliberation. The present example demonstrates only the weighted-sum calculation and its sensitivity to selected weights; it does not demonstrate Win-Win negotiation or a completed stakeholder decision process.

The case context is the March 18, 2018 Uber ATG test-vehicle collision in Tempe, Arizona, documented in NTSB/HAR-19/03 (National Transportation Safety Board, 2019). The vehicle used a developmental automated driving system whose intended capability the NTSB described as SAE Level 4. During testing, however, a human operator was required to monitor the driving environment and intervene when necessary. This arrangement corresponds broadly to human-on-the-loop oversight in RE4HOAI terminology; it does not represent operation without human oversight. At the time of the collision, one operator performed monitoring and event-tagging duties, the vehicle manufacturer's collision-warning and automatic-emergency-braking functions were disabled during automated operation, and the developmental system did not provide emergency braking for collision mitigation.

Alternative A represents the pre-crash baseline. Alternative B is an illustrative configuration informed by documented post-crash changes: two operators with differentiated responsibilities, attentiveness monitoring, enabled Volvo collision-warning and automatic-emergency-braking functions, changes permitting emergency braking for mitigation, and improved prediction of pedestrian crossings outside crosswalks. Alternative C is a hypothetical lean configuration retaining these safeguards and attentiveness monitoring with one operator. The alternatives compare oversight arrangements and supporting safeguards rather than different established SAE automation levels. Although B draws on documented changes, its MCDA scores and comparative outcomes remain author-assigned assumptions.

Scores were assigned retrospectively by the authors with knowledge of the crash outcome. They are illustrative judgments informed by the accident context rather than measured performance estimates. Baseline weights reflect the first author's professional judgment; they were not elicited through a formal stakeholder-weighting exercise. No claim is made that the numerical scores or weights were independently established by the stakeholder interviews.

All criteria use a 1-5 scale, with higher scores indicating a more favorable assumed outcome. The criteria are anticipated crash-risk reduction, hazard-response capability, workload and complacency

management, testing efficiency, and cost favorability. A higher cost-favorability score denotes a less costly assumed configuration.

The numerical scores are hypothetical inputs selected to demonstrate the MCDA procedure, not estimates derived from the accident investigation. Documented differences between configurations inform the illustrative rationale, but do not establish the magnitude of the assigned scores. Other defensible scoring assumptions could produce different totals and rankings.

Configuration	Anticipated crash-risk reduction	Hazard-response capability	Workload and complacency management	Efficiency	Cost favorability	Weighted total
A. Pre-crash baseline	1	1	1	5	5	1.8
B. Configuration informed by post-crash changes	5	5	4	3	2	4.3
C. Hypothetical lean configuration	4	5	3	4	3	3.9

Table 1. Assumed scores for demonstrating the MCDA procedure

Under these assumptions, B ranks above C and A. The ordering follows the assigned scores and weights and does not demonstrate that B would deliver the assumed real-world outcomes.

Weight sensitivity scenarios: Four analyst-selected scenarios were calculated while holding the scores fixed. Weight vectors below follow the same five-criterion order and are shown as percentages. These scenarios illustrate alternative priorities rather than empirically observed stakeholder preferences.

Scenario	Weights (%)	A	B	C	Ranking
Baseline	40/20/20/10/10	1.80	4.30	3.90	B > C > A
Equal weights	20/20/20/20/20	2.60	3.80	3.80	B = C > A
Safety emphasis	60/15/15/5/5	1.40	4.60	3.95	B > C > A
Efficiency and cost emphasis	20/10/10/30/30	3.40	3.40	3.70	C > A = B

Table 2. Sensitivity of weighted totals and rankings

The sensitivity analysis compares four illustrative weighting scenarios while holding the scores fixed. It demonstrates that alternative priorities can change the ranking but does not exhaustively assess possible weights or uncertainty in the scores.

B ranks highest under baseline and safety-emphasis weights, ties with C under equal weights, and is overtaken by C in the efficiency-and-cost-emphasis scenario. The ranking is therefore conditional on the priorities encoded by the weights. The analysis illustrates why a decision rationale should disclose its assumptions and examine sensitivity rather than present one configuration as objectively optimal.

This sensitivity analysis varies weights only. It does not test uncertainty in the scores, estimate crash probabilities or financial costs, or establish implementation compatibility. The weighted sum assumes compensability and additive contributions; it does not establish that a configuration satisfies mandatory safety constraints. Prospectively elicited stakeholder priorities, substantiated performance evidence, and assessment of feasibility would be needed to support an actual engineering decision.

Win-Win negotiation and stakeholder workshops remain candidate deliberation methods within RE4HOAI, but neither is demonstrated nor evaluated here. This example illustrates MCDA calculation and documentation, not a complete negotiation process or validation of the framework.

References

Dubljevic, V., List, G., Milojevich, J., Ajmeri, N., Bauer, W. A., Singh, M. P., Bardaka, E., Birkland, T. A., Edwards, C. H. W., Mayer, R. C., Muntean, I., Powers, T. M., Rakha, H. A., Ricks, V. A., & Samandar, M. S. (2021). Toward a rational and ethical sociotechnical system of autonomous vehicles: A novel application of multi-criteria decision analysis. PLOS ONE, 16(8), e0256224. https://doi.org/10.1371/journal.pone.0256224

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